Checkpoint C - Quality and Residual Risk

2026-11-12

Risk map

Risk	Control	Effort	Residual
Look-ahead leaks into a rule	availability bars; a causality test per event	high	a new reading can repeat a past mistake
Non-determinism across runs	fixed accumulation order; sorted merges; one digest	high	equality across two CPUs is unproven
A run loses an artifact silently	publish only after every hash verifies	medium	a failure during verification is untested in production
A change breaks recorded evidence	versioned contracts; immutable runs	medium	a digest definition change invalidates old bundles
A wrong input is accepted as valid	strict config validation; cross-config checks	medium	a valid config can still be a bad hypothesis

One control is not a test: contract versioning plus the review rule.

One behaviour, with evidence

- **Risk:** look-ahead.
- **Behaviour:** an event appears only at or after the bar on which it became knowable.
- **Expected:** appending later bars does not change any earlier published value.
- **Observed:** the prefix and causality tests pass on the current model suite.

Command: `cargo test -p quantrule-model` (causality and prefix tests).

Limit

Passing causality tests prove the tested readings are causal. They do not prove that a strategy written on top of them is: a strategy is a separate rule and can still read a level before it is confirmed.

Three decisions

- **Review.** Kept numeric model output and rejected labels at the service boundary; labels are a client mapping, so two clients cannot drift.
- **AI use.** AI drafted the architecture documents and the work items. Every claim was checked against code or the gate; cross-machine determinism is explicitly not claimed as observed.

- **Residual risk.** The determinism claim is bounded to one machine until a second one runs the verification procedure. Next: add the scale-invariance test before the client ships.

Risk -> Control -> Evidence -> Review -> Residual: the evidence supports the causality claim for the tested readings, not for a strategy built on them.